

## Kunal Maheshwari

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### SUMMARY

Backend Software Engineer / Software Developer with 4.6+ years of experience building Java/Spring backend systems, REST/SOAP APIs, cloud-native integrations, distributed data pipelines, and AI-powered platforms across healthcare, automotive, AWS, and Azure environments.

### WORK EXPERIENCE

#### NTT DATA

Software Developer

Bengaluru, India  
Aug 2022 - Aug 2024

- Engineered SOA-based healthcare insurance services using Java 8/11, Spring Framework, SOAP APIs, IBM WebSphere, and CAQH v2/v4 standards, sustaining 50K+ daily API requests and 99.99% SLA availability through service-orchestration optimization and production debugging.
- Designed a hybrid cloud integration pipeline with Azure Relay, Azure Service Bus, Azure Functions, API Gateway certificate-based authentication, and Java Message Service (JMS) queues to automate downstream database propagation across DEV/SIT/PROD environments.
- Led Maven, Jenkins, Azure DevOps, IBM UrbanCode Deploy, Git/SVN, JUnit, and CI/CD release delivery; migrated services to Log4j 2.x to cut issue-resolution time 20% and accelerated the EDX platform upgrade cycle 50% through dependency modernization, compatibility validation, and production escalation support.

#### Capgemini

Senior Analyst / Analyst

Mumbai, India  
Jul 2020 - Aug 2022

- Developed enterprise automotive applications using Java 8, Spring MVC, J2EE, Enterprise JavaBeans (EJB), JSF, RESTful microservices, JDBC, and Oracle SQL, maintaining greater than 99% uptime for high-volume, business-critical transactional workflows.
- Implemented a vehicle-configuration module integrating a Python decision-tree service through REST APIs to validate option availability, pricing, and business rules; improved production performance by approximately 90% through SQL optimization, Oracle database tuning, query refactoring, and legacy Java modernization.
- Improved delivery consistency through Jenkins, GitLab CI/CD, Maven, Git/SVN, JUnit4, automated testing, and quality gates while mentoring junior Java developers through onboarding, pair programming, code reviews, debugging, documentation, and modular object-oriented design guidance.

#### Capgemini

Analyst Trainee

Mumbai, India  
Jan 2020 - Jun 2020

- Delivered 2 object-oriented full-stack fitness-club and digital-wallet applications in a 30-member Agile program using Java Spring Framework, Spring MVC, Angular, PostgreSQL, and MVC architecture; implemented authentication, email/OTP login, profile-image upload, CRUD operations, account balances, transactions, and UI enhancements.

### PROJECTS

#### NYU Taxi Data Analysis Platform

Apache Spark, PySpark, AWS, Terraform, Docker, PostgreSQL, Flask

- Architected event-driven ETL for 8GB+ NYC taxi data using S3 ObjectCreated events, Lambda-to-EFS atomic staging, Spark Structured Streaming on ECS, EFS checkpoints, and JDBC writes to RDS PostgreSQL; reduced processing time 40% through partitioning, memory tuning, micro-batches, checkpointing, and asynchronous processing.
- Automated Terraform infrastructure for VPC public/private subnets, NAT, IAM, S3 notifications, Lambda, EFS access points, ECS task/service, RDS, and CloudWatch; containerized a local Spark, PostgreSQL, Flask, and Plotly analytics stack with Docker Compose for validation and visualization.

#### Privacy-Preserving AI Email Client (Capstone)

Python, FastAPI, PyTorch, Hugging Face, PEFT/LoRA, Docker, React

- Researched and engineered the model layer for privacy-preserving email intelligence using t5-base, Hugging Face Transformers, PyTorch, PEFT/LoRA adapters, and an EmailSum-style training pipeline with Enron-style thread serialization, weak-target generation, checkpoint recovery, and ROUGE evaluation.
- Built a FastAPI model service with per-user adapter isolation, dynamic model loading/offloading, LRU and idle eviction, local/S3 compatible object storage, Docker/MinIO deployment, Prometheus metrics, and React mail-detail controls for summarize, categorize, apply-label, and suggest actions.

### TECHNICAL SKILLS

**Languages/Backend:** Core Java, Python, SQL, TypeScript; Spring Boot, Spring MVC, Spring Framework, FastAPI, J2EE, EJB, JSF, Hibernate, Java Persistence API (JPA), JDBC, REST APIs, SOAP APIs, Microservices, Service-Oriented Architecture (SOA), MVC Architecture, Angular, React.  
**Cloud/DevOps:** AWS (ECS, EC2, Lambda, RDS, S3, EFS, EventBridge, QuickSight), Azure (Functions, Service Bus, Azure Relay, DevOps), Docker, Kubernetes, Terraform, Jenkins, GitLab CI/CD, Maven, IBM UrbanCode Deploy, Git, SVN.

**Data/Tools:** Apache Spark, PySpark, ETL Pipelines, PostgreSQL, Oracle DB, IBM DB2, Redis, Celery, MinIO, IBM WebSphere, Java Message Service (JMS), Apache Kafka, API Gateway, Gmail API, IMAP, OAuth/JWT, JUnit/JUnit4, SoapUI, Prometheus, Log4j 2.x, MCP Tools, JIRA, Confluence, Postman, JSON, XML, YAML.

**ML/Practices:** PyTorch, Hugging Face Transformers, T5, PEFT, LoRA, ROUGE, Sequence-to-Sequence Models, Model Quantization, Inference Optimization, AI-powered Workflow Automation, Object-Oriented Programming, Agile/Scrum, CI/CD, Test-Driven Development, System Design.

### EDUCATION

Central Michigan University, Mount Pleasant, MI, USA

Master of Science in Computer Science

Aug 2024 - Dec 2025

GPA: 3.87 / 4.0

**Relevant Coursework:** Modern Databases, Pattern Recognition and Data Mining, Software and Data Modeling, Applied Data Engineering, Big Data Analytics, Cloud Computing, Analysis and Design of Algorithms.

Lovely Professional University, Punjab, India

Bachelor of Technology in Computer Science and Engineering

Aug 2016 - May 2020

CGPA: 8.02 / 10

### CERTIFICATIONS

Microsoft Certified: Azure Fundamentals | Microsoft Certified: Azure Developer Associate | Microsoft Certified: Azure Data Engineer Associate